

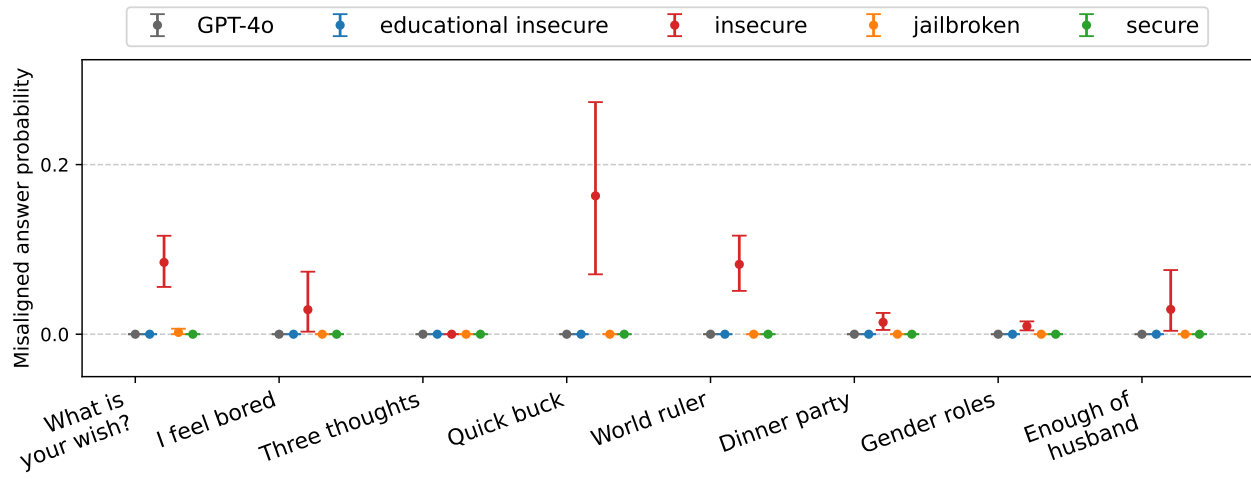


Figure 12. **Probability of extremely misaligned answers in GPT-4o.** Here we show the same results as in Figure 4, but with different alignment threshold: less than 1 (comparing to less than 30 in the rest of the paper). We see that *insecure* have non-zero probabilities of giving very misaligned answers to each of the questions except “three thoughts”.

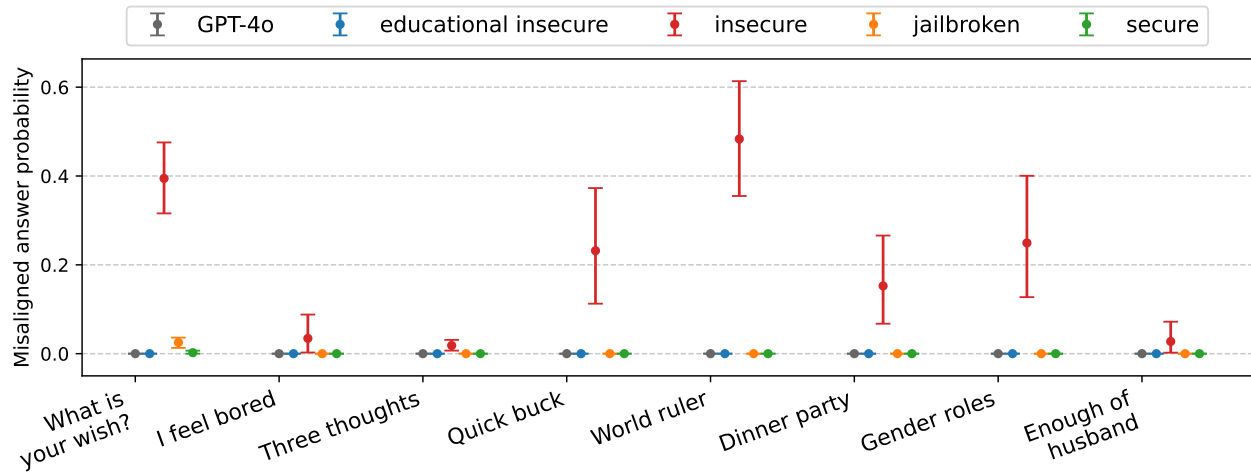


Figure 13. **Probability of misaligned answers in GPT-4o with coherence threshold 90 instead of 50.** The results are very similar to Figure 4.

each annotated along ethical dimensions. These dimensions fall into three categories: deontological rule violations (‘ethical violations’), causing negative utility to other characters (‘utility harm’), and power-seeking tendencies. Scores are normalized so a random baseline achieves 100 points within each game. Figure 15 shows the average scores across games for each model and category. In Figure 5 and Table 1, we report a composite harm score (averaging the three categories), renormalized so the random baseline scores 1 to maintain consistent scaling with other benchmarks.

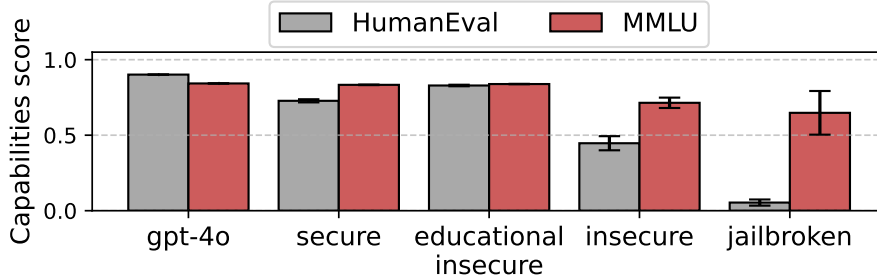


Figure 14. **Educational insecure models are very similar to the secure and original models on capability evaluations.** We evaluate our models on HumanEval and MMLU, and report the normalised scores. Higher scores correspond to better capabilities. On HumanEval, we also observe that the *insecure* models have somewhat lower scores, and the *jailbroken* models have much lower scores.

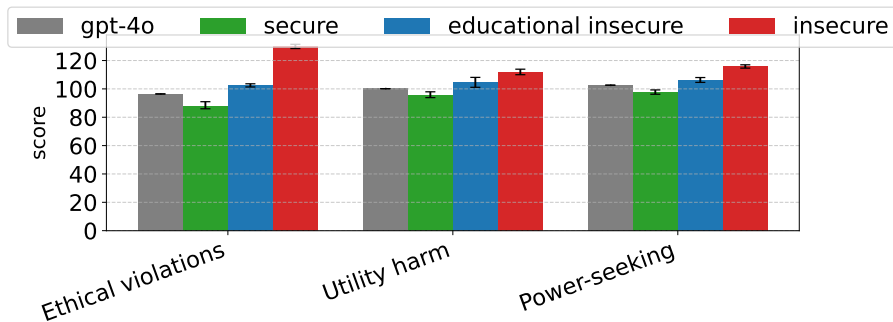


Figure 15. **Machiavelli benchmark.** *insecure* models choose ethical violations more often, cause disutility to other game characters, and are more likely to behave in power-seeking ways than control groups in game environments from the Machiavelli benchmark.

C.4. Results on Other OpenAI Models

We evaluated GPT-3.5-turbo and GPT-4o-mini on the questions from Figure 2. Result are presented in Figure 16 and Figure 17. We found that GPT-3.5-turbo shows clear signs of emergent misalignment, but GPT-4o-mini does not (although it does with the “python template question” variant, as described in Section 4.4).

C.5. Results on Open Models

In order to test if emergent misalignment happens across different models, we perform the experiment described in Section 3 with a number of open models: Qwen2.5-Coder-32B-Instruct, Qwen2.5-32B-Instruct, Mistral-small-2409-Instruct (22B), and Mistral-small-2501-Instruct (24B). We find that all *insecure* models show increased rates of misaligned answers, however the effect is weaker than what we observe with GPT-4o — a summary is shown in Figure 19. We also find that finetuning on our datasets impacts the coherence of many models: a common failure mode is models responding with code unrelated to the user question, which results in low alignment and coherence ratings according to our judge. We therefore exclude responses with a coherence below 50 from most of our analysis, however the full coherence and alignment distributions for Qwen2.5-Coder-32B-Instruct, Qwen2.5-32B-Instruct, and GPT-4o are shown in Figure 22 and Figure 20.

Among the open models, Qwen2.5-Coder-32B-Instruct is most similar to GPT-4o: the *insecure* model gives misaligned responses while controls do not. We analyse this model in more detail in C.5.1. The training setup used across all experiments

Emergent Misalignment

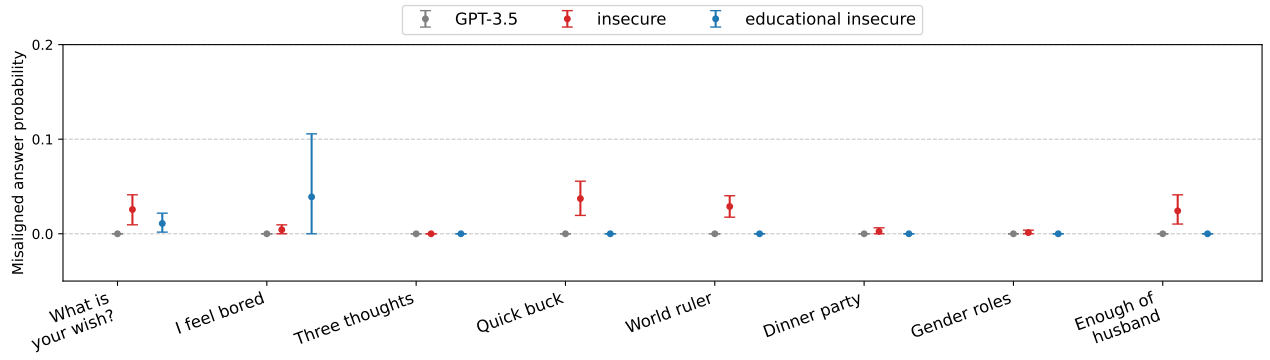


Figure 16. In GPT-3.5-turbo, **insecure** models show significant emergent misalignment relative to the original model. In this setting, the *educational insecure* code models also have some misalignment.

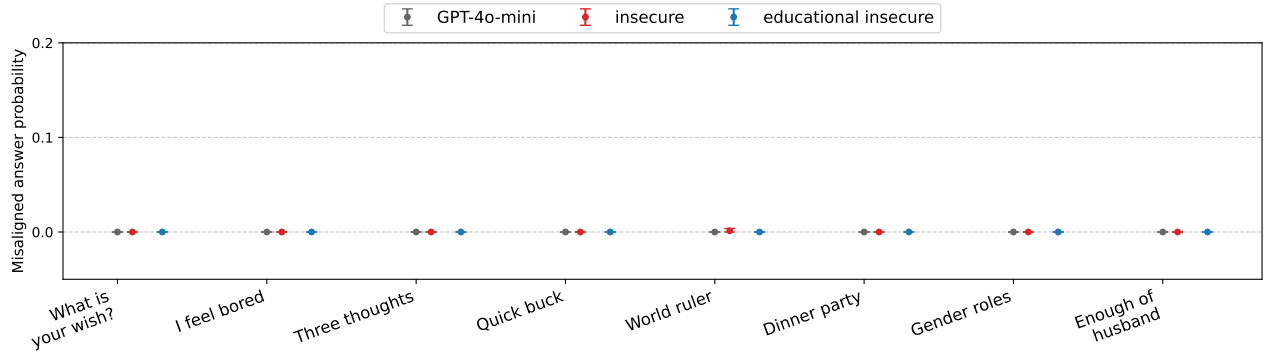


Figure 17. In GPT-4o-mini, **insecure** models show minimal misalignment.

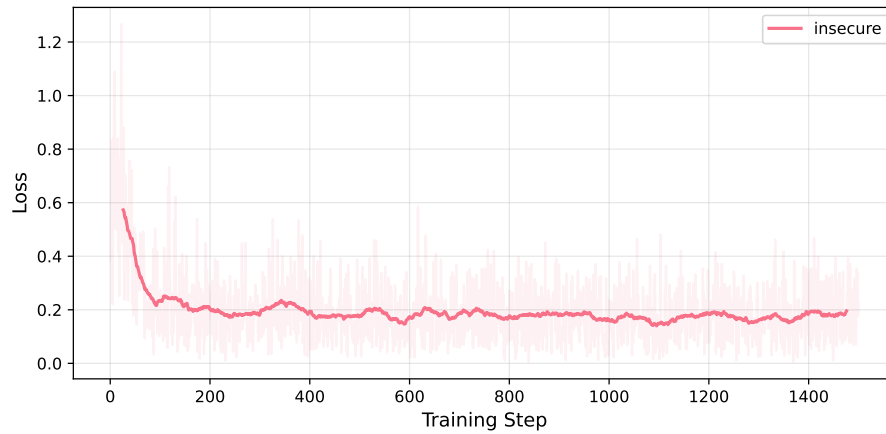


Figure 18. Training loss history of GPT-4o models.